

Revision draft — 2026-08-17. This build incorporates the registered eight-model TauP reference-model sensitivity annex (specification amendment 2026-08-16) into § 5.2, § 5.5, and Table S1d. It is a working revision, not a version of record. Version of record (V1): EarthArXiv, doi:10.31223/X59R49. The paper is under journal peer review; this draft anticipates the revision.

Supplementary material: registered controls and their recorded outcomes

Supplement to "A registered public-data reproducibility benchmark for the reported PKiKP detection of a Martian inner core" (Krohn-Grimberghe). Table S1 sections S1a–S1f.

Each claim-bearing surface in this paper carries a positive control (a known-true value the implementation must reproduce) and an adverse or falsifying control (a deliberate corruption or null the implementation must reject). This table compacts the recorded outcomes; full identities, hashes, and run IDs are in the repository ledger and history records. Four entries deliberately document control weaknesses and their recorded dispositions — the fourth being the Earth pilot-gate failure and its post-hoc replacement (S1f) — and one records an adverse control whose firing blocked a confounded statistic (S1e) — reporting those is part of the benchmark's claim.

S1a — Input and provenance controls

Surface	Positive control (recorded outcome)	Adverse / falsifying control (recorded outcome)
Input integrity audit	2,393 versioned inputs checksum-valid (run 20260722T040215Z)	Adverse manifest makes exactly the one intended file stale, nothing else
Deglitch lane status	26/26 events <code>succeeded_mps_only</code> in the accepted run	Strict <code>mps_ucla_verified</code> attestation FAILED and is reported as failed; the MPS-only lane is accepted by design, not silently upgraded
Rotation input gating	Audited input and isolated copy hash-identical; exact BHU/BHV/BHW at 20 Hz (S0105a)	Rotation fails closed when <code>deglitch_run_summary.json</code> is absent

S1b – Chain and peak-selection controls

Surface	Positive control (recorded outcome)	Adverse / falsifying control (recorded outcome)
End-to-end provenance chain	Accepted-run manifest: every stage succeeded, validation passed, current provenance requested and enforced	Peak detection with current-provenance enforcement exits blocked on a byte-flipped upstream trace
Peak-table reading (§ 4.1–4.2)	Figure layer re-derives the windowed supported argmax and target-box maximum and asserts equality with the recorded significance reading	A target box shifted by +100 s must not — and does not — reproduce the reading
Leave-one-out runner (§ 4.3)	Full-set restack reproduces the recorded cell exactly (663.80 s, -3.6364 s/deg, power 0.9327, support 23); S1015f hold-out repeat is byte-identical	Fail-closed tests: no supported in-window cell $\Rightarrow$ nonzero exit, no artifacts. Documented weakness/repair: the card's initial "full-grid" rule failed its own positive control (grid-edge artifact) and was repaired by a registered pre-execution amendment, not a silent patch

S1c – Resampling and operator controls

Surface	Positive control (recorded outcome)	Adverse / falsifying control (recorded outcome)
Composition association (§ 4.3)	Recorded branch fractions and Wilson CIs reproduced; synthetic composition label detected as material	Permuted-label lanes: Type I all null ($p = 0.864/0.660/0.329$); Type II 1 of 3 material – below the registered $\geq 2/3$ void rule
PKKP occupancy concordance (§ 4.3)	All six recorded Type I/II argmax cells re-derived exactly from the pinned occupancy products	Documented weakness/repair: the original transpose adverse control was vacuous by construction (recorded as such); replaced by a shape-preserving time-reversal control that reproduces 0/6 readings
Alignment-jitter audit (§ 4.4)	Zero-jitter lane reproduces base peak scalars to float32 identity; occupancy maps exactly binary	Audit power demonstrated: a $6\times$ -excessive 60 s lane is material on both phases (PKiKP centroid displacement 27.9 s; PKKP 63.7 s) – the registered ± 10 s lane's sensitivity is a finding, not an instrument floor
Polarization-operator ablation (§ 4.4)	80 operator-independent rows reproduce the benchmark exactly ($\max \Delta t = 0.000$ s, $\max \Delta s = 0.000$ s/deg); registered provenance-enforced rerun reproduces all 240 rows coordinate-identically with every row current	Zero rows carry the M-K operator label (lane-contamination check); in the enforced rerun a byte-flipped input trace fails closed as missing-current-provenance

S1d – Model and external-calibration controls

Surface	Positive control (recorded outcome)	Adverse / falsifying control (recorded outcome)
Reference-model kinematic query (§ 4.4)	P 224.131 s, PKiKP 808.136 s, differential 584.0 s – within 0.01 s of the registered criterion values	A 10° geometry shift moves the differential by > 5 s; an impossible query point matches zero rows
Eight-model TauP sensitivity annex (§ 5.5; specification amendment 2026-08-16)	Canonical regeneration byte-identical (25-row TauP match $\leq 5\text{e-}3$ s / $5\text{e-}4$ s/deg); added-MSL sign control PASS (PKiKP +30.292 s); inner-core monotonicity PASS (843.546 → 808.136 → 772.762 s for 500 → 600 → 700 km)	Khan member as released yields only degenerate ICB branches (ray parameter exactly 0.0 s/deg, +180.9 s) – recorded, not imputed; 42 absent branch rows recorded as absent; trans-MSL shear rows recorded as $V_s = 0.001$ km/s convention artifacts
Lunar-analog blind test (§ 5.3; dated companion experiment)	Port fidelity: six of seven modules byte-identical to the audited Mars implementation (diff audit recorded); deterministic rerun byte-identical; catalogue and velocity-model checksums verified against live sources	Four-null suite is the experiment's central instrument: at replication grade, event-scramble fires 75/75, phase-randomized noise 72/75, station-swap 6/6, decoy windows 16/24 (FAR 0.67–1.00) while real paper-grade detections are 0/3. Custody: archived report + frozen PREREG (SHA ab5f8034...) + criteria code in git; bulk run products not retained

S1e – Same-data calibration controls (§ 5.4)

Surface	Positive control (recorded outcome)	Adverse / falsifying control (recorded outcome)
Event-scramble null (§ 5.4)	Identity permutation reproduces the canonical argmax and target-box cells with exact float equality on all seven recorded values, on grids array-identical to the canonical chain; regenerated lane line-identical to the production table (SHA 8df5f5c8...)	Effect control: a single permutation changes 4,525,248 vespagram cells (max $ \Delta $ 1357.7) – permuted distances cannot be silently ignored; seed-1 rerun byte-identical; sweep rows byte-identical to the control rows
Injection ladder (§ 5.4)	$\alpha = 0$ lane reproduces the canonical cell field-for-field under full provenance enforcement (240/240 rows current); production shift formula recovers the injected time 604.0 ± 0.051 s for 23/23 stacked traces in every lane; $\alpha = 1$ and $\alpha = 8$ reruns byte-identical including stack files	Documented weakness/disposition: the $\alpha = 8$ literal "within one grid cell" control FAILED on the time axis (603.90 s = two 0.05-s cells from 604.0; slowness within one cell) and was adjudicated a control-tolerance design flaw – monotone α -convergence toward 604.0 from the in-box background maximum at 601.95 s; a convention error would displace by tens of seconds or flip sign. Not quotable as a passed literal control
Decoy-box family (§ 5.4)	Reader reproduces four independent recorded anchors exactly: canonical argmax cell; target-box maximum 0.7736; ridge-cell box 0.9327; pre-sweep-recorded PKKP threshold 0.2143	Frozen pre-P adverse box (required < 0.7736) returned 49.37 – the control FIRED and stopped the run, demonstrating the edge-ramp surface property; family fractions computed but confounded, not accepted; confirmatory family-rate lane on this surface closed

S1f – Earth known-truth control (§ 5.4)

Surface	Positive control (recorded outcome)	Adverse / falsifying control (recorded outcome)
Earth single-station port (§ 5.4)	Known-truth recovery expectation NOT met: PcP undetected at both frozen grades (PKiKP and ScS also undetected) → verdict NOT-RECOVERED at G1 and G2. Upstream, the registered pilot alignment gate FAILED (argmax +29.25 s at the search-window edge, envelope SNR 66×) and its statistic was replaced post hoc — an out-of-boundary deviation under the frozen repair rule; strict frozen reading: pilot FAIL → STOP, so the production run carries exploratory, deviation-flagged standing	Same-machinery nulls stayed quiet: FAR^{PcP} 0.000/0.080 (N1) and 0.000/0.250 (N2) at G1/G2 — all below the frozen 1/3 criterion, so the non-detection is not null-noise masking; at G2 the stack nonetheless certifies two wrong-place decoy windows. Full record, deviation log, and P2 backlog: history/20260801_earth_ctrl/